HTTPv2 The docking protocol interface document of facial recognition terminal

- **Communication protocol based on HTTP**
- **Document version 6.0**
- *Release date: June 12, 2024**
- **Modified by Lai Jinjie**

Update Record

Version Number	Modified Date	Explanation

HTTP Protocol Description

- **Supports HTTP 1.1
- **GZIP compression technology**
- **Supports TLS1.2 and TLS1.3**

HTTP API interface

Call process

-The device send a heartbeat packet to the server first, and the server responds with an OK message

- Device sends working parameters to the service
- After the server responds to the heartbeat protection package, it decides the next request to be initiated based on the response content
- **Priority syncParameter > Remote > deletePeople > addPeople**
- After processing the operations required by the server, check for any accumulated un uploaded records
- If there are accumulated records that have not been uploaded, push the records to the server
- The device needs to ensure that the interval between two heartbeat keep alive packets does not exceed the set polling interval.
- The heartbeat interval is 15 seconds. If the first heartbeat packet is sent at 10:00:00 and there is a request to delete or add personnel in the server response, the delete or add personnel interface will be called repeatedly. However, it is necessary to ensure that another heartbeat is sent at 10:00:15.
    - **Processing parameter synchronization, responding to remote operations, deleting personnel, adding personnel, and pushing records between two heartbeat intervals.**

Regarding HTTP Connection Maintenance

The device and server should first establish a connection through a keep alive packet. After the server responds successfully, it should maintain a long HTTP connection until all operations are completed before disconnecting. If there are no operations to be performed, the connection can be immediately disconnected until the next keep alive packet

- When the server supports the HTTP2.0 protocol, priority should be given to using HTTP2.0 for requests
- One device should ensure that it only maintains one HTTP connection with the server to avoid multiple concurrent connections

Exception handling

Network abnormality

- 1.Unable to connect to the server upon startup, failed to send parameter upload request
    - Record parameter upload status as' not uploaded ', communication enters sleep mode, waiting for the next interval time
    - When the interval time is up, send heartbeat packets. If the server does not respond, continue to sleep and wait
    - After the heartbeat packet responds, check the parameter upload status. If it is not successfully sent, send it again, and then continue to respond to the subsequent operations requested by the server (refer to the calling process)
- 2.The device cannot connect to the server during operation
    - Any request should be accompanied by a timeout and response retry - it is recommended to set a timeout of 2 seconds and retry the request 3 times
      After the request fails, enter communication sleep mode and wait for the interval between sending the keep alive packet to arrive
    - Follow up process reference question 1

The server refused the connection

- After the device sends a punch-in record to the server API URL request, the response returned by the server
- If the success field returns a value of 401/403, it indicates no permission. The device should enter request silence and wait for the next keep alive cycle. After the keep alive cycle, send the keep alive first
- When the Status status in the heap is 401/403/400, etc., it is also necessary to wait for the next keep alive packet cycle before sending a keep alive packet to detect the server status.
- Continue executing other API calls only when the server returns an HTTP status of 200 and a Success field of 1

Connect to keep alive

API Device Heartbeat Active Package

Brief description:

- When the device is idle, send a keep alive packet more than [keep alive interval] seconds after the last communication with the server to confirm if there are any unfinished tasks

Request URL:

- http://Server IP:port**/Device/Keepalive**

Request method:

- POST

Content-Type

- application/json

Request Parameters

Field	Type	Explain
SN	string	Device SN
RelayStatus	int	Relay state 0- indicates COM and NC are normally closed 1- indicates COM and NO are normally closed
KeepOpenStatus	int	Normally open 0- indicates normally closed 1- indicates normally open
DoorSensorStatus	int	Door sensor status 0- indicates closed 1- indicates open
LockDoorStatus	int	Door locked status 0- indicates unlocked 1- indicates locked
AlarmStatus	string	Door alarm status An empty string indicates no alarm, otherwise there will be a specific alarm name fire -- fire alarm blacklist -- blacklist alarm anti -- tamper alarm illegal -- unauthorized verification password -- force alarm password openTimeout -- door timeout alarm doorSensor -- door sensor alarm When there are multiple alarms, use commas to separate it, such as fire, blacklist

Example of Request Parameters

```
{
    "SN":"FC-8200H12345678",
    "RelayStatus":  0,
    "KeepOpenStatus":   0,
    "DoorSensorStatus": 0,
    "LockDoorStatus":   0,
    "AlarmStatus":  "",
}
```

Parameter description of return value

Parameter Name	Type	Necessary	Description
Success	int	Yes	1: Success< 401/403 indicates that the server has been connected, but subsequent connections are denied due to the SN not being registered or installed
AddPeople	int	No	>0- indicates that there is a need to add personnel to the device After receiving the device, a request for DevicePass/SelectPassInfo needs to be initiated =0 or when there is no such field, it indicates that no processing is required
DeletePeople	int	No	>Indicates the need to delete personnel from the device After the device receives it, it needs to initiate a request for DevicePass/selectDeleteInfo
SyncParameter	int	No	1- Indicates that there are parameters that need to be set to the device After the device receives them, it needs to initiate a Device/DownloadWorkSetting request
Remote	int	No	1- Indicates that there is a remote operation that needs to be processed. After the device receives it, it needs to initiate a device/setRestart request
UploadWorkParameter	int	No	1- Indicates that the device is required to upload its operating parameters< After receiving the device, it needs to initiate a Device/UnplanadWorkSetting request

Example of return value

```
{
    "Success":0,
    "AddPeople":1,
    "DeletePeople":1,
    "SyncParameter":1,
    "Remote":1,
    "UploadWorkParameter": 1
}

{
    "Success":0
}
```

- **Priority UploadWorkParameter >Remote > SyncParameter> DeletePeople> AddPeople**

Return error code

```
{
    "Success":401   //When returning non-zero values such as 401 and 400. If the device encounters a problem on the platform, it will no longer send punch in records and device parameters to the server. But it will still periodically send keep alive packages
}
/*
When testing the connection of the device again, prompt the text based on the return value
(1) 401 prompt
① Chinese: Connected to the server, but the device is not activated
②English: Connected to server, but device is not activated
(2) 400 prompt
① Chinese: Connected to server, but SN does not comply with platform rules
② Chinese: Connected to server, but SN does not comply with platform rules
(3) 0 prompt
① Chinese: Connected to server, testing completed
②English: Connected to server, testing completed
(4) Other prompts based on message content
*/
```

Device operating parameters

###Device Basic Information SystemInfo

Field	Type	Explain
DeviceSN	string	Serial number
DeviceName	string	Device name
FirmwareVerson	string	Firmware version

Field	Type	Explain
FingerprintVerson	string	Fingerprint algorithm version
FaceVerson	string	Facial algorithm version
Manufacturer	string	Manufacturer
ManufacturerPhone	string	Contact No.
Website	string	Website
ProductionDate	string	Date of manufacture
OEMText	string	OEM custom text, can be filled in with 200 characters
AutoRestart	int	Daily automatic restart function switch
AutoRestartTime	string	Daily automatic restart time, format HH: mm

Working status Status

Field	Type	Explain
RunDays	int	System operation days
FormatCount	int	Format times
WatchDogCount	int	Watchdog reset times
BootTime	int	Startup time Unix timestamp (seconds)
RelayStatus	int	Relay state 0- indicates COM and NC are normally closed 1- indicates COM and NO are normally closed
KeepOpenStatus	int	Normally open 0- indicates normally closed 1- indicates normally open开
DoorSensorStatus	int	Door sensor status 0- indicates closed 1- indicates open
LockDoorStatus	int	Door locked status 0- indicates unlocked 1- indicates locked

Field	Type	Explain
AlarmStatus	string	Door alarm status An empty string indicates no alarm, otherwise there will be a specific alarm name fire -- fire alarm blacklist -- blacklist alarm anti -- tamper alarm illegal -- unauthorized verification password -- force alarm password openTimeout -- door timeout alarm doorSensor -- door sensor alarm When there are multiple alarms, use commas to separate it, such as fire, blacklist

Region and Language Language

Field	Type	Explain
Language	int	Language 1- Chinese< Br>2- English; 3- Traditional Chinese; 4- French; 5- Russian 6- Portuguese; 7- Spanish; 8- Italian language; 9- Japanese 10- Korean; 11- Thai language; 12- Arabic; 13- Portugal 14- Türkiye, 15- Indonesia, 16- Ukraine, 17- Vietnam
SystemTime	int	Device time Unix timestamp (seconds)
UseNTP	int	Enable NTP automatic time calibration 1- Enable; 0-- Disabled
TimeZone	int	Time zone of the device, value range: -12-+14
Volume	int	Volume size (range 0-10)
Voice	int	Voice playback switch 0. No broadcast 1. Broadcast

Human Computer Interaction UI

Field	Type	Explain
DisplayBrightness	int	Screen brightness setting 1-10
MenuPassword	string	Menu password, only number, 4-8 digits or blank
ShowIR	int	Display infrared image on the device 1- Enable; 0-- Disabled
ShowPersonPhoto	int	Display personnel picture after recognition 1-- Enable; 0-- Disabled
PlayPersonName	int	Identify personnel and broadcast the names 1-- Enable; 0-- Disabled
RecognitionButon	int	Before recognition, please click the recognition button 1- Enable; 0-- Disabled

Field	Type	Explain
UnregisteredWarn	int	Unregistered personnel reminder 1-- Enable; 0-- Disabled
ShowPersonName	int	Display the personnel name after recognition? 1- Enable; 0-- Disabled
FillLight	int	Supplement light mode: 0: Normally closed; 1: Normally open; 2: Automatic;
UseQRCode	int	QR code recognition switch 1- enabled; 0-- Disabled

data storage Storage

Field	Type	Explain
RecordAutoCycle	int	Record full cycle 1- Record full cycle, 0- Record full not cycle, waiting for cleaning
SaveUnregistered	int	Save unregistered personnel records, 0: do not store, 1: store. Unregistered personnel refer to individuals who were not registered in the system, or card numbers that were not registered in the system
SaveRecordPicture	int	Save on-site image 0. do not save; 1. Save
PeopleStorageInfo	class	Personnel storage details
RecordStorageInfo	class	Record storage details

- **PeopleStorageInfo**

```
{
"Person": {"Max“: 5000, "Current":0 },//Personnel storage capacity Max maximum capacity; Current current storage quantity
"Face": {"Max“: 5000, "Current":0 },//Face storage capacity
"Card": {"Max“: 5000, "Current":0 },//Card storage capacity
"Fingerprint": {"Max“: 5000, "Current":0 },//Fingerprint storage capacity
"PalmVein": {"Max“: 5000, "Current":0 },//Palm print storage capacity
"Pasword": {"Max“: 5000, "Current":0 },//Password storage capacity
"Admin": {"Max“: 5000, "Current":0 }//Administrator storage capacity
}
```

- **RecordStorageInfo**

```
{
"VerifyRecord": {"Max": 5000, "Current":0 },//Storage capacity for entry and
exit records
"DoorRecord": {"Max": 5000, "Current":0 },//Storage capacity for door sensor
"SystemRecord": {"Max": 5000, "Current":0 },//Storage capacity for system
records
"RecordPhoto": {"Max": 5000, "Current":0 }//Storage capacity for live picture
}
```

Face Recognition

Field	Type	Explain
FaceIR	int	Liveness detection, 1 on, 0 off
FaceIRThreshold	int	Liveness detection threshold 1-99
FaceDistance	int	Identification distance 1- near range (0.2-0.5 meters); 2- Middle distance (0.2-1.5 meters); 3-- far distance (0.2-1.5 meters or more)
FaceThreshold	int	The face recognition threshold is 1-99. The recognition threshold more larger, the accuracy more higher
FPComparison	int	Fingerprint comparison threshold value range: 1-100
FaceMask	int	Mask detection
FaceMaskThreshold	int	The mask recognition threshold is 1-99. The recognition threshold more larger, the accuracy more higher

BodyTemperature

Field	Type	Explain
UseBodyTemperature	Int	Temperature measurement mode switch. 0: Non temperature measurement mode 1: Temperature measurement mode
UseFahrenheitDisplay	int	Turn on Fahrenheit temperature display, 1: on, 0: off
TemperatureCompensate	float	Temperature compensation value -10.0 -- +10.0
TemperatureAlarmThreshold	float	Example of temperature alarm threshold 37.5
TemperatureDisplay	int	Whether to display body temperature '0' -prohibit displaying body temperature; 1-- Display body temperature info

NetworkServer

Fields	Type	Explain
UseTCPClient	Int	Use TCPClient to connect to server 1- Enable; 0-- Disabled;
UseUDPClient	Int	Use UDPClient to connect to server 1- Enable; 0-- Disabled;
ServerAddress	string	Server Address TCP or UDP Protocol Server Address
ServerPort	int	Server port number
KeepaliveTime	int	kEEP alive package interval time 1-65535 seconds
UseHTTPClient	int	Enable HTTP Client protocol 1- Enable; 0- Disabled;
HTTPClient_ServerAddr	string	HTTP protocol server address
HTTPClient_KeepaliveTime	int	Time interval for keep alive packets in HTTP protocol
HTTPClient_UseGZIP	int	Use GZIP compression when making HTTP protocol requests? 0- Not used; 1- Used
HTTPClient_ProtocolType	int	Protocol type of HTTPClient 100 --- HTTPv1 200 ---HTTPv2
UseMQTTClient	int	Whether to start MQTTClient protocol 1- Enable; 0- Disabled;
UseMQTTSSL	int	Enable MQTT SSL Secure Socket 1- Enable; 0- Disabled;
MQTTServerAddr	string	MQTT server address www.abc.com
MQTTPort	int	MQTT server port number
MQTTLoginName	string	Login username in MQTT protocol
MQTTLoginPassword	string	Login password in MQTT protocol
MQTTPublishTopic	string	Topic used by devices to send data in MQTT protocol
MQTTSubscribeTopic	string	Topics that devices in the MQTT protocol need to subscribe to when receiving data

Fields	Type	Explain
MQTT_KeepaliveTime	int	The keep alive packet interval time of MQTT protocol
MQTT_UseGZIP	int	Does MQTT use GZIP compression? 0- Not used; 1- Used
UseWebsocketClient	int	Whether to start Websocket Client protocol 1-Enable; 0-- Disabled;
WebsocketClient_ServerAddr	string	Websocket protocol server address ws://192.168.1.1/websocket or wss://192.168.1.1/websocket
WebsocketClient_KeepaliveTime	int	The keep alive packet interval time of Websocket Client protocol
WebsocketClient_UseGZIP	int	Does Websocket use GZIP compression? 0-Not used; 1- Used
WebsocketClient_ProtocolType	int	Protocol types of Websocket
UseYZW	int	Do you want to start the HTTPClient protocol for Yunzhu platform 1- Enable; 0-- Disabled;
YZWAddr	string	Yunzhu platform protocol server address

Machine network parameters

Field	Type	Explain
UseWired	int	Wired network switch, 1: on, 0: off
WiredDHCP	int	Wired network automatic IP, 1: on, 0: off
WiredIP	string	Wired network IP address ("//192.168.0.110")
WiredIPMask	string	Wired network subnet mask ("//255.255.255.0")
WiredGteway	string	Wired network gateway ("//192.168.0.1")
WiredDNS	string	Dns(“//192.168.0.1”)
WiredMAC	string	Wired network MAC address
UseWIFI	int	Wireless network switch, 1: on, 0: off
WIFIAPName	string	Wireless network account
WIFIAPPassword	string	Wireless network password

Field	Type	Explain
WIFIMAC	string	Wireless network MAC address
WIFIDHCP	int	Wireless network automatic IP, 1: on, 0: off
WIFIIP	string	Wireless network IP address ("//192.168.0.110")
WIFIIPMask	string	Wireless network subnet mask(“//255.255.255.0”)
WIFIGteway	string	Wireless network gateway(“//192.168.0.1”)
WIFIDNS	string	Wireless Network Dns(“//192.168.0.1”)
UseWebPage	int	Web page management switch, 1: on, 0: off
HTTPPort	int	Web management page port number, 1-65534
HTTPSPort	int	Web management page port number, 1-65534
WebPageUseSSL	int	Enable SSL on device web page. SSL certificate is self signed using OpenSSL
UseUDP	int	UDP port switch, 1: on, 0: off
UDPPort	int	UDP port number (used by UDP protocol)
ConnectPassword	string	UDP protocol communication password 32 characters
UseTelnet	int	Linux Telnet function switch, 1: on, 0: off
TelnetPort	int	Telnet port number
UseRTSP	int	RTSP video stream, 1: on, 0: off
RTSPPort	int	RTSP port number
RTSPUser	string	RTSP username
RTSPPassword	string	RTSP password

###Door access parameters

Field	Type	Required	Explain
CardBytes	int	Yes	Card number byte; 3. 4, 8; 0- indicates disabling card reading

Field	Type	Required	Explain
AccessType	int	Yes	Entry and exit type 0, entry; 1, exit
WgFormat	int	Yes	WG Output format 26 / 34/66
WGContent	int	Yes	WG Output content: 1- User ID; 2-- Card number
ReleaseTime	int	Yes	Opening holding time 0-65535(s). 0 is 0.5 seconds
DelayOpenDoorTime	int	Yes	Delay unlocking time 0-65535 (s). 0 is prohibition
FreeOpen	int	Yes	Without verification to open door 1-enabled; 0-- Disabled
OpenInterval	int	Yes	Repeat recognition interval 0-disabled; 1-65535（ms)
OpenInterval_SaveRecord	int	Yes	Repeat interval record storage setting 0, do not save; 1, save
Relay	int	Yes	The relay support bistability?``1 is support,0 is not support
ShortMessage	string	Yes	Short messages after legal verification
VerificationType	int	Yes	Verification method< br/>1. Standard mode; 2. Face/fingerprint/palm print/card+password< br/>2. Face/fingerprint/palm print/card+password< br/>3. card+face/fingerprint/palm print/password< br/>4. Multi person attendance 5. People and ID comparison< br/>6. card+face/fingerprint/palm print+password< br/>7. card+fingerprint/palm print+face recognition< br/>8. Fingerprint/palm print+face+password< br/>9. Fingerprint+palm print+face< br/>10. Palm print+face; 11. Fingerprint+facial recognition< br/>12. Only use palm print; 13. Only use fingerprints; 14. Only use card; 15. Only use password< br/>16. People and ID comparison to open the door and auto-registered (ID card+face+auto-registered);

Field	Type	Required	Explain
OverdueRemind	int	Yes	Permission expiration prompt 1--Enable; 0--Disabled
OverdueRemind_Day	int	Yes	Permission expiration prompt Validity threshold, minimum remaining valid days. If the number of days is below this threshold, it will prompt that the validity period is about to expire. The value range is 1-255. 0- indicates closed.
TimingOpen	int	Yes	Timing normally open function 1--Enable； 0--Disabled
TimingOpen_mode	int	Yes	Timing normally open. Automatic opening mode: ` `Timing normally open. Automatic opening mode: 1. After passing legal authentication, it can be normally open within a specified period of time. 2. Those marked as normally open privilege in the authorization can be normally open after passing authentication within a specified period of time. 3. Automatic switch, the door will automatically open and close when the time is up
TimingOpen_timegroup	object	Yes	Timing Normally opened. The time zone of normally opened using the weekly period structure
TimingLocked	int	Yes	Timing locked function 1--Enabled； 0--Disabled
TimingLocked_timegroup	object	Yes	Timing locked.The time zone of timing locked using the weekly period structure
VisitorRootPassword	string	Yes	Root password of visitor
MultiPerson	int	Yes	Multi person combination to open the door, number of people; 1-50；

- VerificationType

```
  1. Standard mode default value
2. Face/fingerprint/palm print/card+password
3. Card+face/fingerprint/palm print/password
4. Multi person attendance
5. People and ID Comparison
6. Card+face/fingerprint/palm print+password
7. Card+fingerprint/palm print+face recognition
8. Fingerprint/palm print+face+password
9. Fingerprint+palm print+face
10. Palm print+face
11. Fingerprint+Face
12. Only palm print
13. Only fingerprint
14. Only card
15. Only password
16. People and ID comparison to open door + auto registered（identification
+face +auto registered）
```

- **Weekly time zone format**

```
{

        Week1:"00:00-23:59", //Monday
        Week2:"00:00-23:59",
        Week3:"00:00-23:59",
        Week4:"00:00-23:59",
        Week5:"00:00-23:59",
        week6:"00:00-23:59",
        week7:"00:00-23:59" //Sunday
}
```

  - Week1 is Monday
  - Week2 is Tuesday
  - Week3 is Wednesday
  - Week4 is Thursday
  - Week5 is Friday
  - Week6 is Saturday
  - Week7 is Sunday
  - Each week field represents the time one setting for a day
  - You can set up eight sub time zone in a day with the format of start time-end time/start time-end time/.....

```
"01:00-01:59/02:00-02:59/03:00-03:59/04:00-04:59/05:00-05:59/06:00-06:59/07:00-
07:59/08:00-08:59"
The above string defines 8 sub time zone in a day. A maximum of 8 sub time zone
can be defined per day
"01:00-01:59/02:00--02:59"
The above string defines 2 sub time zone in a day, and the other 6 time zone are
empty and invalid.
```

If the day of the week is empty and no time zone is set, this field can be omitted

```
  {
        //At this time, only week 1 and week 7 are defined, and other time zone
are empty
        Week1:"01:00-02:00",
        Week7:"03:00-04:00"
  }
```

Parameters of elevator function

Field	Type	Required	Explain
UseElevator	int	Yes	Elevator function switch, 1: on, 0: off
ElevatorPorts	[] Object array	Yes	Elevator Port Object array defines elevator port list

Elevator Port Object

Field	Type	Required	Explain
Num	int	Yes	Elevator port number 1-64
RelayType	int	Yes	Elevator relay (COM&NO normally closed, COM&NC normally closed) ` ` Value range: 1. COM&NC normally closed (default value); 2. COM&NO normally closed
ReleaseTime	int	Yes	The maximum output duration during unlocking is 65535 seconds. 0 represents 0.5 seconds
TimingOpen	obj	Yes	Timing normally open function structure

- timingOpen Timing normally open function structure

Field	Type	Required	Explain
Use	int	Yes	Function switch,0--Disabled； 1--Enabled
Open	int	Yes	Automatic opening mode: 1. After passing the legal authentication, the door can be normally opened within a specified period of time. 2. Those marked as normally open privilege in the authorization can be normally opened after passing the authentication within the specified period of time. 3. Automatic switch, the door will automatically open and close when the time is up
Timegroup	obj	Yes	Normally open time zone, using weekly time zone structure

Alarm parameters

Field	Type	Required	Explain
FireAlarm	int	Yes	Fire alarm,0,Diabled； 1,Enabled
DoorLongOpenAlarm	int	Yes	Opening timeout alarm switch,1:Enabled,0:Diabled
DoorLongOpenTime	int	Yes	Opening timeout, if the door is opened for more than this time, an alarm will be triggered 1-65535 （s）
DoorSensorAlarm	int	Yes	Door sensor alarm,0,Diabled；1,Enabled
DoorSensorAlarmTimegroup	class	Yes	Door sensor alarm and non alarm time zone, weekly time zone format
BlacklistAlarm	int	Yes	Blacklist alarm,0,Diabled；1,Enabled
AntiDisassemblyAlarm	int	Yes	Tamper alarm function switch,0,Diabled； 1,Enabled
IllegalVerificationAlarm	int	Yes	Illegal verification alarm function,0,Diabled； 1,Enabled
IllegalVerificationAlarmLimit	int	Yes	Illegal verification alarm function-Number of illegal authentication attempts,1-255
UseUserCloseAlarm	int	Yes	Allow users verification to remove the alarm switch,0,Diabled；1,Enabled
PasswordAlarm	int	Yes	Forced alarm password function,0,Diabled； 1,Enabled

Field	Type	Required	Explain
PasswordAlarm_Password	string	Yes	Forced alarm password, entering this password will trigger an alarm. Passwords only support numbers and can contain 0.
PasswordAlarm_Mode	string	Yes	The working mode when the forced alarm occurs: 1- Do not open the door, alarm output: 2- Open the door, alarm output: Lock the door, alarm, only can be unlocked by software``

###Device opening time zone Timegroup

Filed	Type	Required	Explain
TimeGroups	[] Object array	Yes	Device opening time zone

Format description of device opening time zone

```
//Parameter
TimeGroups:[
    {
        Num:1,
        Week1:"00:00-23:59", //Monday
        Week2:"00:00-23:59",
        Week3:"00:00-23:59",
        Week4:"00:00-23:59",
        Week5:"00:00-23:59",
        Week6:"00:00-23:59",
        Week7:"00:00-23:59" //Sunday
    },
    {
        Num:2,
        Week1:"01:00-01:59/02:00-02:59/01:00-01:59/02:00-02:59",
        ..
        Week7:"00:00-00:00"
    },
    ......
    {
        Num:64,
        Week1:"00:00-00:00",
        ..
        Week7:"00:00-00:00"
    }
]
```

-The maximum num of the opening time zone of device is 64, indicating that the device has a maximum of 64 sets of opening time zone

- week1 Indicate Monday
- week2 Indicate Tuesday
- week3 Indicate Wednesday
- week4 Indicate Thursday
- week5 Indicate Friday
- week5 Indicate Saturday
- week7 Indicate Sunday

-Each week field represents the time zone setting for a day

-You can set up 8 sub time zone in a day with the format of start time end time/start time-end time/.....

```
"01:00-01:59/02:00-02:59/03:00-03:59/04:00-04:59/05:00-05:59/06:00-06:59/07:00-07:59/08:00-08:59"
The above string defines 8 sub time zone in a day. A maximum of 8 sub time zone can be defined per day
"01:00-01:59/02:00-02:59"
The above string defines 2 sub time zone in a day, and the other 6 time zone are empty and invalid.
```

- If the day of the week is empty and no time zone is set, this field can be omitted

```
{
        Num:64,  //At this time, only week 1 and week 7 are defined, and other time periods are empty
        Week1:"01:00-02:00",
        Week7:"03:00-04:00"
    }
```

Holiday of device

Field	Type	Required	Explain
Holidays	[]	Yes	Holiday

```
[
 {"Num":1,"Date":"2020-10-01","Type":1,"Cycle":1},
 {"Num":2,"Date":"2020-10-02","Type":2,"Cycle":0},
...
]
```

Use object array format during holidays, with each object containing two fields, num and date.
The device currently supports 360 sets of holidays
On holidays, it is prohibited to open doors (permission can be set for holiday passage)

*Field Description of *Holiday Object**

Field	Type	Required	Explain
Num	int	Yes	Number of holidays, used when binding personnel permissions

Field	Type	Required	Explain
Date	string	Yes	Holiday Date Year-Month-Day Example: 2020-10-01
Type	int	No	Holiday control range,1--all day； 2--Morning 00:00-12:00; 3--Afternoon(12:00-23:59),default value is 1;
Cycle	int	No	Cycle annually?,1--cycle annually； 0--Non cycle; The default value is 0;

AlarmClock

Up to 24 alarms

Field	Type	Required	Description
AlarmClocks	[]	Yes	Alarm Clock

```
[
 {"Num":1,"Clock":"12:00","Times":10},
 {"Num":2,"Clock":"13:00","Times":10},
 {"Num":3,"Clock":"14:00","Times":10},
...
]
```

The alarm uses an object array format, each object containing three fields, Num, Clock, Times. The device currently supports 24 sets of alarms

Alarm Object Field Description

Field	Type	Required	Decription
Num	int	Yes	The serial number of the alarm clock, with a value range of 1-24
Clock	int	Yes	Alarm time HHmm format example: 1230 represents 12:30 pm
Times	int	No	Alarm duration, with a value range of 0-255 in seconds

Device Function List FunctionList

```
{
    "FunctionList" :{
        //Temperature detection
        "BodyTemperature": true,
        //Fingerprint
        "Fingerprint": true,
        //Palm Print
```

```
        "Palmvein": true,
        //Face Recognition
        "Face": true,
        //QR Code
        "QRCode": true,
        //Mask Detection
        "FaceMask": true,
        //Helmet Detection
        "SafetyHelmet": true,
        //Elevator
        "Lift": true,
        //Alarm Clock
        "AlarmClock": true,
        //excel Export and Import
        "ExcelFile": true,
        //zip Import
        "ZipFile": true,
        //Quantity of time zone
        "TimeGreoup": true,
        //Wireless network
        "WIFI": true,
        //HTTPClient v1
        "HTTPClient_V1": true,
        //HTTPClient v2
        "HTTPClient_V2": true,
        //MQTT
        "MQTT": true,
        //Cloud Building Network Platform
        "YZW": true,
        //Websocket V1
        "Websocket_V1": true,
        //Websocket V2
        "Websocket_V2": true
    }
}
```

API-Upload current device parameters

Brief description:

- Send once when the device is turned on
- When the device properties change, they will be uploaded again

Request URL:

- http://Server IP:port**/Device/UploadWorkSetting**

Request method:

- POST

Content-Type

- application/json

Request parameters:

- **Example of Request Parameters**


```
{
        "DeviceSN": "FC-8200H12345675",
         "FireAlarm": 0,
         "DoorLongOpenAlarm": 0,
         "DoorLongOpenTime": 0,
         "DoorSensorAlarm": 0,
         "DoorSensorAlarmTimegroup": {
             "Week1": "",
             "Week2": "",
             "Week3": "",
             "Week4": "",
             "Week5": "",
             "Week6": "",
             "Week7": ""
         },
         "BlacklistAlarm": 1,
         "AntiDisassemblyAlarm": 1,
         "IllegalVerificationAlarm": 0,
         "IllegalVerificationAlarmLimit": 30,
         "UseUserCloseAlarm": 1,
         "PasswordAlarm": 0,
         "PasswordAlarm_Password": "",
         "PasswordAlarm_Mode": 1,
         "AlarmClocks": [
             {
                 "Num": 1,
                 "Clock": "00:00",
                 "Times": 0
             },
             {
                 "Num": 24,
                 "Clock": "00:00",
                 "Times": 0
             }
         ],
         "UseBodyTemperature": 1,
         "UseFahrenheitDisplay": 1,
         "TemperatureCompensate": 0,
         "TemperatureAlarmThreshold": 37.29999,
         "TemperatureDisplay": 1,
         "CardBytes": 3,
         "AccessType": 0,
         "WgFormat": 26,
         "WGContent": 1,
         "ReleaseTime": 3,
         "FreeOpen": 0,
         "OpenInterval": 2,
         "OpenInterval_SaveRecord": 0,
         "Relay": 0,
         "ShortMessage": "",
         "VerificationType": 1,
```

```
        "OverdueRemind": 1,
        "OverdueRemind_Day": 3,
        "TimingOpen": 0,
        "TimingOpen_mode": 1,
        "TimingOpen_timegroup": {
            "Week1": "",
            "Week2": "",
            "Week3": "",
            "Week4": "",
            "Week5": "",
            "Week6": "",
            "Week7": ""
        },
        "TimingLocked": 0,
        "TimingLocked_timegroup": {
            "Week1": "",
            "Week2": "",
            "Week3": "",
            "Week4": "",
            "Week5": "",
            "Week6": "",
            "Week7": ""
        },
        "VisitorRootPassword": "",
        "MultiPerson": 1,
        "UseElevator": 0,
        "ElevatorPorts": [
            {
                "Num": 1,
                "RelayType": 2,
                "ReleaseTime": 2,
                "TimingOpen": {
                    "Use": 0,
                    "Open": 3,
                    "Timegroup": {
                        "Week1": "",
                        "Week2": "",
                        "Week3": "",
                        "Week4": "",
                        "Week5": "",
                        "Week6": "",
                        "Week7": ""
                    }
                }
            },
            {
                "Num": 64,
                "RelayType": 2,
                "ReleaseTime": 2,
                "TimingOpen": {
                    "Use": 0,
                    "Open": 3,
                    "Timegroup": {
                        "Week1": "",
                        "Week7": ""
```

```
            }
        }
    }
],
"FaceIR": 1,
"FaceIRThreshold": 5,
"FaceDistance": 3,
"FaceThreshold": 58,
"FPComparison": 80,
"FaceMask": 0,
"FaceMaskThreshold": 65,
"Holidays": [],
"Language": 1,
"SystemTime": 1718421632,
"UseNTP": 1,
"TimeZone": 8,
"Volume": 6,
"Voice": 1,
"ConnectPassword": "",
"UseWired": 1,
"WiredDHCP": 0,
"WiredIP": "192.168.1.103",
"WiredIPMask": "255.255.255.0",
"WiredGteway": "192.168.1.1",
"WiredDNS": "192.168.1.1",
"WiredMAC": "7E-87-16-C1-E6-51",
"UseWIFI": 0,
"WIFIDHCP": 0,
"WIFIIP": "192.168.1.150",
"WIFIIPMask": "255.255.255.0",
"WIFIGteway": "192.168.1.1",
"WIFIDNS": "192.168.1.1",
"WIFIMAC": "34-7D-E4-2D-63-3B",
"WIFIAPName": "",
"WIFIAPPassword": "",
"UseWebPage": 1,
"HTTPPort": 80,
"HTTPSPort": 443,
"WebPageUseSSL": 1,
"UseUDP": 1,
"UDPPort": 8101,
"UseTelnet": 1,
"TelnetPort": 23,
"UseRTSP": 1,
"RTSPPort": 554,
"RTSPUser": "admin",
"RTSPPassword": "12345678",
"UseTCPClient": 0,
"UseUDPClient": 0,
"ServerAddress": "47.92.31.75",
"ServerPort": 9003,
"KeepaliveTime": 30,
"UseHTTPClient": 0,
"HTTPClient_ProtocolType": 100,
"HTTPClient_ServerAddr": "http://192.168.1.100",
```

```
        "HTTPClient_KeepaliveTime": 30,
        "HTTPClient_UseGZIP": 0,
        "UseMQTTClient": 0,
        "UseMQTTSSL": 0,
        "MQTTServerAddr": "192.168.1.100",
        "MQTTPort": 0,
        "MQTTLoginName": "",
        "MQTTLoginPassword": "",
        "MQTTPublishTopic": "",
        "MQTTSubscribeTopic": "",
        "MQTT_KeepaliveTime": 30,
        "MQTT_UseGZIP": 0,
        "UseWebsocketClient": 0,
        "WebsocketClient_ProtocolType": 100,
        "WebsocketClient_ServerAddr": "ws://192.168.1.100/ws",
        "WebsocketClient_UseGZIP": 0,
        "WebsocketClient_KeepaliveTime": 30,
        "UseYZW": 0,
        "YZWAddr": "http://192.168.1.10",
        "RunDays": 0,
        "FormatCount": 0,
        "WatchDogCount": 7,
        "BootTime": 1718418194,
        "RelayStatus": 0,
        "KeepOpenStatus": 0,
        "DoorSensorStatus": 0,
        "LockDoorStatus": 0,
        "AlarmStatus": "\"\"",
        "RecordAutoCycle": 0,
        "SaveUnregistered": 1,
        "SaveRecordPicture": 1,
        "PeopleStorageInfo": {
            "Person": {
                "Max": 20000,
                "Current": 1
            },
            "Face": {
                "Max": 20000,
                "Current": 1
            },
            "Card": {
                "Max": 20000,
                "Current": 0
            },
            "Fingerprint": {
                "Max": 0,
                "Current": 0
            },
            "PalmVein": {
                "Max": 10000,
                "Current": 0
            },
            "Pasword": {
                "Max": 20000,
                "Current": 0
```

```
        },
        "Admin": {
            "Max": 5,
            "Current": 0
        }
    },
    "RecordStorageInfo": {
        "VerifyRecord": {
            "Max": 1000000,
            "Current": 4
        },
        "DoorRecord": {
            "Max": 10000,
            "Current": 4
        },
        "SystemRecord": {
            "Max": 10000,
            "Current": 7
        },
        "RecordPhoto": {
            "Max": 10000,
            "Current": 4
        }
    },
    "DeviceSN": "FC-8190H24052799",
    "DeviceName": "",
    "FirmwareVerson": "8.46",
    "FingerprintVerson": "-",
    "FaceVerson": "6.01",
    "Manufacturer": "",
    "ManufacturerPhone": "",
    "Website": "",
    "ProductionDate": "2024-06-15",
    "OEMText": "",
    "AutoRestart": 0,
    "AutoRestartTime": "02:00",
    "TimeGroups": [
        {
            "Num": 1,
            "Week1": "00:00-23:59",
            "Week2": "00:00-23:59",
            "Week3": "00:00-23:59",
            "Week4": "00:00-23:59",
            "Week5": "00:00-23:59",
            "Week6": "00:00-23:59",
            "Week7": "00:00-23:59"
        },
        {
            "Num": 64,
            "Week1": "",
            "Week7": ""
        }
    ],
    "DisplayBrightness": 6,
    "MenuPassword": "0000",
```

```
        "ShowIR": 0,
        "ShowPersonPhoto": 1,
        "PlayPersonName": 1,
        "RecognitionButon": 0,
        "UnregisteredWarn": 0,
        "ShowPersonName": 1,
        "FillLight": 2,
        "FunctionList": {
            "BodyTemperature":  true,
            "Fingerprint":  false,
            "Palmvein": true,
            "Face": true,
            "QRCode":   true,
            "FaceMask": true,
            "SafetyHelmet": false,
            "Lift": true,
            "AlarmClock":   true,
            "ExcelFile":    false,
            "ZipFile":  false,
            "TimeGreoup":   64,
            "WIFI": true,
            "HTTPClient_V1":    true,
            "HTTPClient_V2":    true,
            "MQTT": true,
            "YZW":  true,
            "Websocket_V1": true,
            "Websocket_V2": true
        }
}
```

Return parameter description

Parameter Name	Type	Nnecessary	Description
Success	int	Yes	1 indicates successful operation; != 1, it is necessary to indicate the error description in the Message
Message	string	No	Error message description

Return value example

```
{
    "Success":1
}
```

API-The device actively obtains working parameters

Brief description:

When the device sends a heartbeat packet and the SyncParameter field in the server's response packet is set to 1, the device will request this interface

Request URL:

- http://Server IP:port**/Device/DownloadWorkSetting**

Request method:

- POST

Content-Type

- application/json

Request parameter description

Parameter Name	Type	Nnecessary	Description
SN	string	Yes	Device ID

Example of Request Parameters

```
{
    "SN":"FC-8200H12345678"
}
```

Return parameter description

Field	Type	Required	Explain
Success	int	Yes	1-- indicates success Other are error codes, and the error content needs to be indicated in the Message
Message	string	No	Error message description, please refer to Chapter 3 for details
All Modified Parameters			

Return value example

```
{
       "Success": 0,
       "DeviceSN": "FC-8200H12345675",
        "FireAlarm": 0,
        "DoorLongOpenAlarm": 0,
        "DoorLongOpenTime": 0,
        "DoorSensorAlarm": 0,
        "DoorSensorAlarmTimegroup": {
            "Week1": "",
            "Week2": "",
            "Week3": "",
            "Week4": "",
            "Week5": "",
            "Week6": "",
            "Week7": ""
```

```
},
"BlacklistAlarm": 1,
"AntiDisassemblyAlarm": 1,
"IllegalVerificationAlarm": 0,
"IllegalVerificationAlarmLimit": 30,
"UseUserCloseAlarm": 1,
"PasswordAlarm": 0,
"PasswordAlarm_Password": "",
"PasswordAlarm_Mode": 1,
"AlarmClocks": [
    {
        "Num": 1,
        "Clock": "00:00",
        "Times": 0
    },
    {
        "Num": 24,
        "Clock": "00:00",
        "Times": 0
    }
],
"UseBodyTemperature": 1,
"UseFahrenheitDisplay": 1,
"TemperatureCompensate": 0,
"TemperatureAlarmThreshold": 37.29999,
"TemperatureDisplay": 1,
"CardBytes": 3,
"AccessType": 0,
"WgFormat": 26,
"WGContent": 1,
"ReleaseTime": 3,
"FreeOpen": 0,
"OpenInterval": 2,
"OpenInterval_SaveRecord": 0,
"Relay": 0,
"ShortMessage": "",
"VerificationType": 1,
"OverdueRemind": 1,
"OverdueRemind_Day": 3,
"TimingOpen": 0,
"TimingOpen_mode": 1,
"TimingOpen_timegroup": {
    "Week1": "",
    "Week2": "",
    "Week3": "",
    "Week4": "",
    "Week5": "",
    "Week6": "",
    "Week7": ""
},
"TimingLocked": 0,
"TimingLocked_timegroup": {
    "Week1": "",
    "Week2": "",
    "Week3": "",
```

```
            "Week4": "",
            "Week5": "",
            "Week6": "",
            "Week7": ""
        },
        "VisitorRootPassword": "",
        "MultiPerson": 1,
        "UseElevator": 0,
        "ElevatorPorts": [
            {
                "Num": 1,
                "RelayType": 2,
                "ReleaseTime": 2,
                "TimingOpen": {
                    "Use": 0,
                    "Open": 3,
                    "Timegroup": {
                        "Week1": "",
                        "Week2": "",
                        "Week3": "",
                        "Week4": "",
                        "Week5": "",
                        "Week6": "",
                        "Week7": ""
                    }
                }
            },
            {
                "Num": 64,
                "RelayType": 2,
                "ReleaseTime": 2,
                "TimingOpen": {
                    "Use": 0,
                    "Open": 3,
                    "Timegroup": {
                        "Week1": "",
                        "Week7": ""
                    }
                }
            }
        ],
        "FaceIR": 1,
        "FaceIRThreshold": 5,
        "FaceDistance": 3,
        "FaceThreshold": 58,
        "FPComparison": 80,
        "FaceMask": 0,
        "FaceMaskThreshold": 65,
        "Holidays": [],
        "Language": 1,
        "SystemTime": 1718421632,
        "UseNTP": 1,
        "TimeZone": 8,
        "Volume": 6,
        "Voice": 1,
```

```
        "ConnectPassword": "",
        "UseWired": 1,
        "WiredDHCP": 0,
        "WiredIP": "192.168.1.103",
        "WiredIPMask": "255.255.255.0",
        "WiredGteway": "192.168.1.1",
        "WiredDNS": "192.168.1.1",
        "WiredMAC": "7E-87-16-C1-E6-51",
        "UseWIFI": 0,
        "WIFIDHCP": 0,
        "WIFIIP": "192.168.1.150",
        "WIFIIPMask": "255.255.255.0",
        "WIFIGteway": "192.168.1.1",
        "WIFIDNS": "192.168.1.1",
        "WIFIMAC": "34-7D-E4-2D-63-3B",
        "WIFIAPName": "",
        "WIFIAPPassword": "",
        "UseWebPage": 1,
        "HTTPPort": 80,
        "HTTPSPort": 443,
        "WebPageUseSSL": 1,
        "UseUDP": 1,
        "UDPPort": 8101,
        "UseTelnet": 1,
        "TelnetPort": 23,
        "UseRTSP": 1,
        "RTSPPort": 554,
        "RTSPUser": "admin",
        "RTSPPassword": "12345678",
        "UseTCPClient": 0,
        "UseUDPClient": 0,
        "ServerAddress": "47.92.31.75",
        "ServerPort": 9003,
        "KeepaliveTime": 30,
        "UseHTTPClient": 0,
        "HTTPClient_ProtocolType": 100,
        "HTTPClient_ServerAddr": "http://192.168.1.100",
        "HTTPClient_KeepaliveTime": 30,
        "HTTPClient_UseGZIP": 0,
        "UseMQTTClient": 0,
        "UseMQTTSSL": 0,
        "MQTTServerAddr": "192.168.1.100",
        "MQTTPort": 0,
        "MQTTLoginName": "",
        "MQTTLoginPassword": "",
        "MQTTPublishTopic": "",
        "MQTTSubscribeTopic": "",
        "MQTT_KeepaliveTime": 30,
        "MQTT_UseGZIP": 0,
        "UseWebsocketClient": 0,
        "WebsocketClient_ProtocolType": 100,
        "WebsocketClient_ServerAddr": "ws://192.168.1.100/ws",
        "WebsocketClient_UseGZIP": 0,
        "WebsocketClient_KeepaliveTime": 30,
        "UseYZW": 0,
```

```
        "YZWAddr": "http://192.168.1.10",
        "RecordAutoCycle": 0,
        "SaveUnregistered": 1,
        "SaveRecordPicture": 1,
        "DeviceName": "",
        "Manufacturer": "",
        "ManufacturerPhone": "",
        "Website": "",
        "ProductionDate": "2024-06-15",
        "OEMText": "",
        "AutoRestart": 0,
        "AutoRestartTime": "02:00",
        "TimeGroups": [
            {
                "Num": 1,
                "Week1": "00:00-23:59",
                "Week2": "00:00-23:59",
                "Week3": "00:00-23:59",
                "Week4": "00:00-23:59",
                "Week5": "00:00-23:59",
                "Week6": "00:00-23:59",
                "Week7": "00:00-23:59"
            },
            {
                "Num": 64,
                "Week1": "",
                "Week7": ""
            }
        ],
        "DisplayBrightness": 6,
        "MenuPassword": "0000",
        "ShowIR": 0,
        "ShowPersonPhoto": 1,
        "PlayPersonName": 1,
        "RecognitionButon": 0,
        "UnregisteredWarn": 0,
        "ShowPersonName": 1,
        "FillLight": 2
}
```


Remote control

API-Remote operation command

Brief description:

- When the Remote value in the heartbeat packet return is 1, the device immediately requests this interface to obtain remote operation commands

Request URL:

- http://Server IP:port**/Device/RemoteCommand**

Request method:

- POST

Content-Type

- application/json

Request parameters:

Field	Type	Length	Required	Explain
SN	string	30	Yes	Device ID

Example of Request Parameters

```
{
    "SN":"FC-8200H12345678"
}
```

Return value parameter description

Parameter Name	Type	Necessary	Description
Success	int	Yes	1-- indicates success Other are error codes, and the error content needs to be indicated in the Message; 0: Success
Message	string	No	Error message description, please refer to Chapter 3 for details
Restart	int	No	Remote restart:``0:no restart，1:restart
Recover	int	No	Factory reset``0:Normal，1:Factory reset
Opendoor	int	No	Remote door opening command` 0:Not processed, 1--Turn on relay; 2--Keep the door open; 3--Close the door (release normally open); `4--Lock the door;5--Unlock the door
Closealarm	int	No	Close alarm command``0:Not processed, 1:Close all current alarms and record them
RepostRecord	int	No	Re-upload records``0:Not processed, 1:Mark all uploaded records as not uploaded and resend them
PushAllPeople	int	No	Request to upload all stored personnel lists to the server At this point, the device calls the API [/People/PushPeople] to send the personnel list --Not processed, 1--All personnel need to be uploaded.

Parameter Name	Type	Necessary	Description
QueryPeople	[uint]	No	Request to upload personnel with the specified user ID to the server. At this point, the device calls the API [/People/PushPeople] to send a list of personnel. The type is an array
ClearRecord	int	No	Delete all records; 0-- Not processed 1--Delete all records

Return value example

```
//Remote door opening
{
    "Success":0,
    "Opendoor":1
}

//Query designated personnel
{
    "Success":0,
    "QueryPeople":[1,2,3,4,5,6]
}
```

#Personnel management

API-Obtain Personnel Authorization Information

Brief description:

--When the heartbeat is kept alive and the server return value with a field for obtaining personnel and this value is 1, the device immediately initiates this request and repeats the request

-Stop request condition: 1. The server returns Success:1, but the personnel list is empty; 2. Success==0

Request URL:

- http://Server IP:port**/People/DownloadPeopleList**

Request Method:

- POST

Content-Type

- application/json

Request parameters

Field	Type	Length	Required	Description
SN	string	30	Yes	Device ID
Limit	int	10	Yes	The maximum number of personnel returned on each request is 1000, and the device sets this value based on its own processing capacity

Example of Request Parameters

```
{
    "SN":"FC-8200H12345678",
    "Limit": 100
}
```

Return value parameter description

Parameter Name	Type	Necessary	Description
Success	int	Yes	1--indicates success Other are error codes, and the error content needs to be indicated in the Message; 0: Success
Message	string	No	Error message description, please refer to Chapter 3 for details
PeopleCount	int	Yes	The number of personnel returned this time
PeopleList	Array	Yes	An array containing personnel permission information structure

PeopleJson Personnel data format

Parameter Name	Type	Neccessary	Description
UserID	string	Yes	User ID (maximum number 4294967295 type UINT32)
Name	string	No	Personnel name (characters<64 bits)
Job	string	No	Position (characters<64 bits)
Department	string	No	Department (characters<64 bits)
IdentityCard	string	No	Identification card number (can be blank) (characters<64 digits)
Attachment	string	No	Other personnel information (characters<200)

Parameter Name	Type	Neccessary	Description
Photo	string		Photo information, starting with http or https means using a URL address, otherwise it means using base64
PhotoMD5	string	No	MD5 HEX string format for photos (characters=32 digits)
PhotoLen	int	No	The maximum length of the photo supported is 400KB
Password	string	No	Password, only number, length: (0/4-8)
CardNum	string	No	Card number (digits, maximum value 18446744407909551615 type UINT62)
QRCode	string	No	Personnel QR code information (characters<128)
AccessType	int	No	Role 0--Ordinary personnel; 1--Administrator; 2-- Blacklist
ExpirationDate	uint32	No	Permission expiration date Unix timestamp in seconds 0 indicates an indefinite period Maximum indicates December 31, 2099
OpenTimes	int	No	Door opening times 0-65535; 65535-- indicates no restrictions, 0--indicates no passage allowed
KeepOpen	int	No	Normally opened card?, 1--Yes;0--No
Timegroup	int	No	Opening time zone group 1-64; 0- indicates restricted
Holidays	string	No	Holiday restrictions comma separated: 1, 2, 3, 4, 5
Elevators	string	No	Elevator port permission array comma separated: 1, 2, 3, 4, 5
FaceFeature	string	No	Facial feature codes starting with http or https indicate the use of a URL address, otherwise they indicate the use of base64. Download the feature code from a file containing the content of base64
FaceFeatureMD5	string	No	MD5 value HEX string format of facial feature code
Fingerprints	[]	No	Fingerprint object

Parameter Name	Type	Neccessary	Description
Palmveins	[]	No	Palm print object

Characters refer to bytes, with one Chinese character occupying 3-4 bytes (encoded in UTF-8) and one English character occupying 1 byte

Holidays Holidays

- An empty string or no field indicates no holiday restrictions
- For specific restricted holiday numbers, comma separated: 1, 2, 3, 4, 5
- *Number indicates that it is subject to all holiday restrictions

- **Fingerprint Fingerprint Object**

Serial Number	Type	Required	Description
Num	int	Yes	Fingerprint index number
Data	string	Yes	The fingerprint signature code starting with http or https indicates the use of a URL address, otherwise it indicates the use of base64. Download the signature code from a file, and the content of the downloaded file is based on base64
MD5	string	No	MD5 value of feature code in HEX string format

```
[ //Structural Example
    {
        Num: 1,
        Data: "http://abc.com/fp/1.dat",
        MD5: "abcdefg"
    },
    {
        Num: 2,
        Data: "abcdefgrtygnfhgfjhk ... jhkghjkgj==",
        MD5: "abcdefg"
    }
]
```

- **Palmvein palm print object**

Serial Number	Type	Required	Description
Num	int	Yes	Palm print index number

Serial Number	Type	Required	Description
Data	string	Yes	The palm print feature code starting with http or https indicates the use of a URL address, otherwise it indicates the use of base64. Download the feature code from a file, and the content of the downloaded file is based on base64
MD5	string	No	MD5 value of feature code in HEX string format

```
[ //Structural Example
    {
        Num: 1,
        Data: "http://abc.com/palm/1.dat",
        MD5: "abcdefg"
    },
    {
        Num: 2,
        Data: "abcdefgrtygnfhgfjhk ... jhkghjkgj==",
        MD5: "abcdefg"
    }
]
```

- **Elevator Elevator Authority**

```
 //Represents this person only has access permission on the 1st-5th floor of elevator
 [
     1,2,3,4,5
 ]
 //Represents this person does not have elevator permission
 [

 ]
 //Represents this person only has access permission on the 10th floor of elevator
 [
     10
 ]
```

Return parameter example

```
{
    "Success": 0,
    "Count": 1,
    "PeopleList": [
        {
            "UserID": "3",
            "Name": "888888",
            "Job": "Development",
```

```
            "Department": "Sales Department",
            "IdentityCard": "",
            "Attachment": "",
            "Photo": "http://abc.com/photo/1.jpg",
            "PhotoMD5": "613D870CA99EDF074BEE4387BAB09070",
            "PhotoLen": 55020,
            "Password": "2222",
            "CardNum": "6666",
            "AccessType": 0,
            "ExpirationDate": 0,
            "OpenTimes": 65535,
            "KeepOpen": 1,
            "Timegroup": 6,
            "Holidays": "1,3,9,10,11,17,21,25,27,30",
            "Elevators": "1,2,3,4,5",
            "FaceFeature": "http://abc.com/face/1.dat",
            "FaceFeatureMD5": "613D870CA99EDF074BEE4387BAB09070",
            "Fingerprints": [
                {
                    Num: 1,
                    Data: "http://abc.com/fp/1.dat",
                    MD5: "abcdefg"
                }
            ],
            "Palmveins": [
                {
                    Num: 1,
                    Data: "http://abc.com/palm/1.dat",
                    MD5: "abcdefg"
                }
            ]
        },
        {
            "UserID": "444",
            "Name": "444",
            ...
            "Fingerprints": [],
            "Palmveins": []
        }
    ]
}
```

API-Feedback to obtain personnel storage results

Brief description:

- After getting a batch of people from the server to import, call this interface to return the import results

Request URL:

- http://Server IP:port**/People/DownloadPeopleListResult**

Request method:

- POST

Content-Type

- application/json

Request parameters:

Field	Type	Length	Required	Description
SN	string	30	Yes	Device ID
SuccessCount	int	20	Yes	Number of successful imports
FailCount	int	20	Yes	Number of failed imports
FailList	[]		Yes	Reason code for import failure (see error code for specific description)

The structure of failure information in the array 'fail Employee Id'

Parameter Name	Type	Necessary	Description
UserID	string	Yes	User ID (maximum number 4294967295 type UINT32)
ErrorCode	int	Yes	Error Code
RepeatID	string	Yes	Duplicate user ID, this field indicates this person is repeating with which person
ErrMsg	string	No	Error Description

Example of Request Parameters

```
{
    "SN":"FC-8200H12345678",
    "SuccessCount":16,
    "FailCount":1,
    "FailCount":[
        {
            "UserID":xxx,
            "ErrorCode":20,
            "RepeatID":123,
             "ErrMsg" : ""

        },{
            "UserID":”xxx”,
            "UserID":1,
             "ErrMsg" : ""
    ]
}
```

*errorCode Personnel failure error code**

Return value parameter description

Parameter Name	Type	Necessary	Description
Success	int	Yes	1--indicates success Other are error codes, and the error content needs to be indicated in the Message; 0: Success
Message	string	No	Error message description, please refer to Chapter 3 for details

Return value example

```
{
    "success":0
}
```

image-20210616205111411

##API - Obtain the personnel to be deleted**

Brief description:

- After sending the heartbeat packet, the server calls back when there are personnel that need to be deleted, and cycle callback
- Stop cycle condition 1, Success:1,and deleteInfo is an empty array or does not have this field 2, Success: 0

Request URL:

- http://Server IP:port**/People/DeletePeopleList**

Request method:

- POST

Content-Type

- application/json

Request parameters:

Field	Type	Length	Required	Explain
SN	string	30	Yes	Device ID
Limit	int	10	Yes	Limit on the number of personnel which returned on each request (default 50 if not carrying, maximum 1000)

Example of Request Parameters

```
{
    "SN":"FC-8200H12345678"
}
```

Return value parameter description

Parameter Name	Type	Necessary	Description
Success	int	Yes	1--indicates success Other are error codes, and the error content needs to be indicated in the Message; 0: Success
Message	string	No	Error message description, please refer to Chapter 3 for details
DeleteAll	int	No	1: Clear all personnel information 0: Delete by specified user number
DeleteCount	int	Yes	Number of personnel to be deleted
DeleteList	[]	No	When DeleteAll=0, this parameter contains the user ID that needs to be deleted

Return value example

```
{
    "Success":0,
    "DeleteAll":0,
    "DeleteList":[
        1,2,3,4,5
    ]
}
```

API-Feedback on the operation results of deleting personnel

Brief description:

- After getting a batch of people from the server to import, call this interface to return the import results

Request URL:

Request method:

- POST

Content-Type

- application/json

Request parameters:

Field	Type	Length	Required	Explain
SN	string	30	Yes	Device ID

Field	Type	Length	Required	Explain
DeleteCount	int	20	Yes	Delete Quantity
DeleteAll	int	20	Yes	1 All personnel have been deleted; 0 Only delete specified personnel
DeleteList	[]		Yes	Deleted personnel number

Example of Request Parameters

```
{
    "SN":"FC-8200H12345678",
    "DeleteCount":5,
    "DeleteAll":0,
    "DeletList":[1,2,3,4,5]
}
```

Parameter Name	Type	Neccessary	Description
Success	int	Yes	1--indicates success Other are error codes, and the error content needs to be indicated in the Message ; 0: Success
Message	string	No	Error message description, please refer to Chapter 3 for details

Return value example

```
{
    "Success":0
}
```

API-Push Personnel Information

Brief description:

- When personnel are added or modified on the device, the changed information will be uploaded to the cloud platform
- When the server requests to upload specified personnel information, push the specified personnel information to the platform
- When the server requests to upload all personnel information, push all personnel information to the platform

Request URL:

- http://Server IP:port**/People/PushPeople**

Request method:

- POST

Content-Type

- multipart/form-data

Request Paramters:

Field	Type	Length	Required	Explain
SN	string	30	Yes	Device ID
PushType	int	1	Yes	Types of personnel changes in device 1--Add New； 2--Update； 3--Delete； 4--Query；
Detail	class		No	Personnel details, if personnel do not exist, this field is not available
Photo	file		No	When there are personnel photos, photo files will be uploaded
UserID	uint32		Yes	Personnel ID The personnel ID for this push

Request parameter examples with personnel photos, fingerprint feature codes, and palm print feature codes

```
POST /note/insertNoteFace HTTP/1.1
Accept: */*
Host: localhost:5000
Accept-Encoding: gzip, deflate, br
Connection: keep-alive
Content-Type: multipart/form-data; boundary=-------------------------
-506873351428002157394455
Content-Length: 17842

----------------------------506873351428002157394455
Content-Disposition: form-data; name="SN"

FC-8200H12345678
----------------------------506873351428002157394455
Content-Disposition: form-data; name="PushType"

1
----------------------------506873351428002157394455
Content-Disposition: form-data; name="Detail"
Content-Encoding: gzip
//The following content needs to be compressed with gzip before transmission.
JSON formatted website https://www.json.cn/
        {
            "UserID": "3",
            "Name": "888888",
            "Job": "Development",
```

```
            "Department": "Sales Department",
            "IdentityCard": "",
            "Attachment": "",
            "Photo": "http://abc.com/photo/1.jpg",
            "PhotoMD5": "613D870CA99EDF074BEE4387BAB09070",
            "PhotoLen": 55020,
            "Password": "2222",
            "CardNum": "6666",
            "AccessType": 0,
            "ExpirationDate": 0,
            "OpenTimes": 65535,
            "KeepOpen": 1,
            "Timegroup": 6,
            "Holidays": "1,3,9,10,11,17,21,25,27,30",
            "Elevators": "1,2,3,4,5",
            "FaceFeature": "http://abc.com/face/1.dat",
            "FaceFeatureMD5": "613D870CA99EDF074BEE4387BAB09070",
            "Fingerprints": [
                {
                    Num: 1,
                    Data: "abcdefgrtygnfhgfjhk ... jhkghjkgj==",
                    MD5: "abcdefg"
                }
            ],
            "Palmveins": [
                {
                    Num: 1,
                    Data: "abcdefgrtygnfhgfjhk ... jhkghjkgj==",
                    MD5: "abcdefg"
                }
            ]
        }
----------------------------506873351428002157394455
Content-Disposition: form-data; name="Photo"; filename="Photo.jpg"
Content-Type: image/jpeg

*****jpeg file binary content*****
----------------------------506873351428002157394455--
```

Return value parameter description

Parameter Name	Type	Neccessary	Description
Success	int	Yes	1-- indicates success Other are error codes, and the error content needs to be indicated in the Message; 0: Success
Message	string	No	Error message description, please refer to Chapter 3 for details

Return value example

```
{
    "Success":0
}
```

Record Management

API-Upload punch in Record

Brief description:

- Use this interface when there are new punch-in records in the device
- This interface only supports uploading one punch-in record at a time

Request URL:

- http://Server IP:port**/Record/UploadIdentifyRecord**

Request method:

- POST

Content-Type

- multipart/form-data

Parameters:

Field	Type	Length	Required	Explain
SN	string	30	Yes	Device SN
RecordDetail	string	1000	Yes	Record details, JSON string
Photo	file	50KB	No	This parameter is only required when there are images in the record

RecordDetail Field Description

Field	Type	Length	Required	Explain
RecordID	long	10	Yes	Serial number of record
RecordType	int	5	Yes	Event type
RecordDate	int64	20	Yes	Punch in time Unix timestamp
UserID	string	20	No	User ID
Name	string	30	No	Personnel Name
IdentityCard	string	5	No	Identity card
Job	string	1	No	Position
Department	string	1	No	Department

Field	Type	Length	Required	Explain
CardNum	string	20	No	Card No.
QRCode	string	128	No	QR Code
IsEntry	int	1	No	Is it entry? 1 indicates entry, 0 indicates exit
BodyTemp	int	5	No	Human body temperature measurement needs to be divided by 10
PhotoLen	int	10	No	Image file length 0 indicates no image

RecordType Event type

Value	Explain
1	Card Verification
2	Fingerprint Verification
3	Facial Verification
4	Card + Fingerprint
5	Face + Fingerprint
6	Card + Face
7	Card + Password
8	Face + Password
9	Fingerprint + Password
10	Password verification user number and password
11	Card + Fingerprint + Password
12	Card+ Face + Password
13	Fingerprint + Face + Password
14	Card + Fingerprint + Face
15	Repeated Verification
16	Expired Validity Period
17	Opening Time Zone Expired
18	Cannot open the door during holidays
19	Unregistered User

Value	Explain
20	Detected Locked
21	The number of valid times has been exhausted
22	Verification while locked, prohibit opening the door
23	Lost Reported Card
24	Blacklist Card
25	Open the door without verification -- when pressing the fingerprint, the user number is 0, and when swiping the card, the user number is the card number
26	Prohibit card swiping verification -- When card swiping is disabled in the [Permission Authentication Method]
27	Prohibit fingerprint verification -- [Permission authentication method] When fingerprint is disabled
28	Controller Expired
29	Verified Passed - Validity period is ready to expire
30	Abnormal body temperature, refusal to enter
31	Visitor password to open the door
32	Scanning dynamic QR code to open the door
33	Add user to the device menu
34	Modify user in the device menu
35	Delete user from the device menu
36	Palm Print Recognition
37	Card + Palm Print+ Face
38	Palm Print + Password
39	Card + Palm Print
40	Face + Palm Print
41	Card + Palm Print + Password
42	Palm Print + Face + Password
43	Fingerprint + Palm Print+ Face
44	Combination verification --waiting for the next person
45	Combination Verification Failed
46	Combination Verification Successful

Value	Explain
47	People and ID Comparison
48	Card Not Registered
49	Unregistered QR Code

RecordDetail Parameter examples

```
{
    "RecordID": 120,
    "RecordType":   3,
    "RecordDate":   1718616771,
    "UserID":   "1",
    "Name": "1",
    "IdentityCard": "",
    "Job":  "",
    "Department":   "",
    "CardNum":  "0",
    "QRCode":   "",
    "IsEntry":  1,
    "BodyTemp": 0,
    "PhotoLen": 40363
}
```

** Request Example**

```
POST /note/insertNoteFace HTTP/1.1
Accept: */*
Host: localhost:5000
Accept-Encoding: gzip, deflate, br
Connection: keep-alive
Content-Type: multipart/form-data; boundary=-------------------------
-506873351428002157394455
Content-Length: 17842
----------------------------506873351428002157394455
Content-Disposition: form-data; name="SN"

 FC-8380T12345678
----------------------------506873351428002157394455
Content-Disposition: form-data; name="recordJson"
Content-Encoding: gzip
Content-Type: application/octet-stream

//When request compression is enabled, the following content needs to be
compressed by gzip before transmission when uploading.
{ "RecordID":120,"RecordType":3,"RecordDate":1718616771,
"UserID":"1","Name":"1","IdentityCard":"","Job":"","Department":"","CardNum":"0"
,"QRCode":"","IsEntry":1,"BodyTemp":0,"PhotoLen":40363}
----------------------------506873351428002157394455
Content-Disposition: form-data; name="pic"; filename="Postman_file.jpg"
Content-Type: image/jpeg
```

```
*****jpeg file binary content*****
----------------------------506873351428002157394455--
```

####* **Return Example**

```
{
    "Success": 0
}
```

Return error code

```
{
    "Success":401   //Returning 401 indicates that the device is not authorized and will no longer send punch in records. The device parameters will be sent to the server. But it will still periodically send keep alive packages
}
//The device returns 401, indicating that the record was not successfully uploaded
```

Request content compression

- If the recordJson field in 'fromdata' is too long, you can use the content compression option and include Content Encoding: gzip in the paragraph

  ```
  Content-Disposition: form-data; name="recordJson"
  Content-Encoding: gzip

    ****Compressed binary content****
  ```

  Return Parameter Description

Parameter Name	Type	Necessary	Description
Success	int	Yes	1-- indicates success Other are error codes, and the error content needs to be indicated in the Message; 0: Success

API-Upload System Records

Brief description:

- Use this interface when there are new system records in the device
- This interface will batch upload system records

Request URL:

- http://Server IP:port/**Record/UploadSystemRecord**

Request method:

- POST

Content-Type

- application/json

Parameter:

Field	Type	Length	Required	Explain
SN	string	30	Yes	Device SN
RecordType	int	1	Yes	Record Type
Records	【】	1000	Yes	Record List

Record Field Description

Field	Type	Length	Required	Explain
RecordID	long	10	Yes	Record number
RecordType	int	5	Yes	Event type 1- Door Sensor Record; 2-- System Record
RecordDate	int64	20	Yes	Punch in time Unix timestamp

RecordType Field Description

Value	Explain
1	Door Sensor Record
2	System Record

Value	Explain
1	Door Sensor-Open Door
2	Door Sensor-Close Door
3	Enter the door sensor alarm detection state
4	Exit the door sensor alarm detection state
5	The door is not closed properly
6	Use the button to open the door
7	When the button is pressed to open the door, the door is already locked

Value	Explain
8	The controller has expired when use the button to open the door

Value	Explain
1	Software Open Door
2	Software Close Door
3	Software Normally Opened
4	The controller automatically enters normally open mode
5	The controller automatically close the door
6	Press and hold the exit button to enters normally opened mode
7	Press and hold the exit button to enters normally closed mode
8	Software Locked
9	Software Removed Locked
10	Controller timing locked--automatic locking at specified time
11	Controller timing locked--automatic remove locking at specified time
12	Alarm--Locked
13	Alarm--Removed Locked
14	Illegal authentication Alarm
15	Door Sensor Alarm
16	Forced Alarm
17	Opening Time Out Alarm
18	Blacklist Alarm
19	Fire Alarm
20	Tamper Alarm
21	Illegal Authentication Alarm Removed
22	Door Sensor Alarm Removed
23	Forced Alarm Removed
24	Timing Opening Timeout Alarm Removed
25	Blacklist Alarm Removed

Value	Explain
26	Fire Alarm Removed
27	Tamper Alarm Removed
28	System Powered On
29	System Error Reset (Watchdog)
30	Device Formatting Record
31	Card Reader Reversed Connection
32	The card reader circuit is not properly connected
33	Unrecognized card reader
34	The network cable has been disconnected
35	The network cable has been inserted
36	WIFI Connected
37	WIFI Disconnected
38	Bluetooth Door Opening
39	Call the Roll Timeout
40	Clear all personnel from the device menu
41	Backup personnel to USB drive in the device menu
42	Import personnel from USB drive in the device menu
43	Remote door opening for indoor unit
44	Delete all records
45	Delete all personnel

RecordDetail Parameter examples

```
[
    {
        "RecordID":    1,
        "RecordType":  1,
        "RecordDate":  1718616771
    },
    {
        "RecordID":    2,
        "RecordType":  1,
        "RecordDate":  1718616772
    }
]
```


Parameter examples


```
{
    "SN":  "FC-8380T12345678",
    "RecordType":  1,
    "Records":  [
        {
            "RecordID":    1,
            "RecordType":  1,
            "RecordDate":  1718616771
        },
        {
            "RecordID":    2,
            "RecordType":  1,
            "RecordDate":  1718616772
        }
    ]
}
```


Return Example


```
{
    "Success": 0
}
```


Return error code

```
{
    "Success":401   //Returning 401 indicates that the device is not authorized and will no longer send punch in records. The device parameters will be sent to the server. But it will still periodically send keep alive packages
}
//The device returns 401, indicating that the record was not successfully uploaded this time
```

Return parameter description

Parameter Name	Type	Necessary	description
Success	int	Yes	1- indicates success Other are error codes, and the error content needs to be indicated in the Message; 0: Success